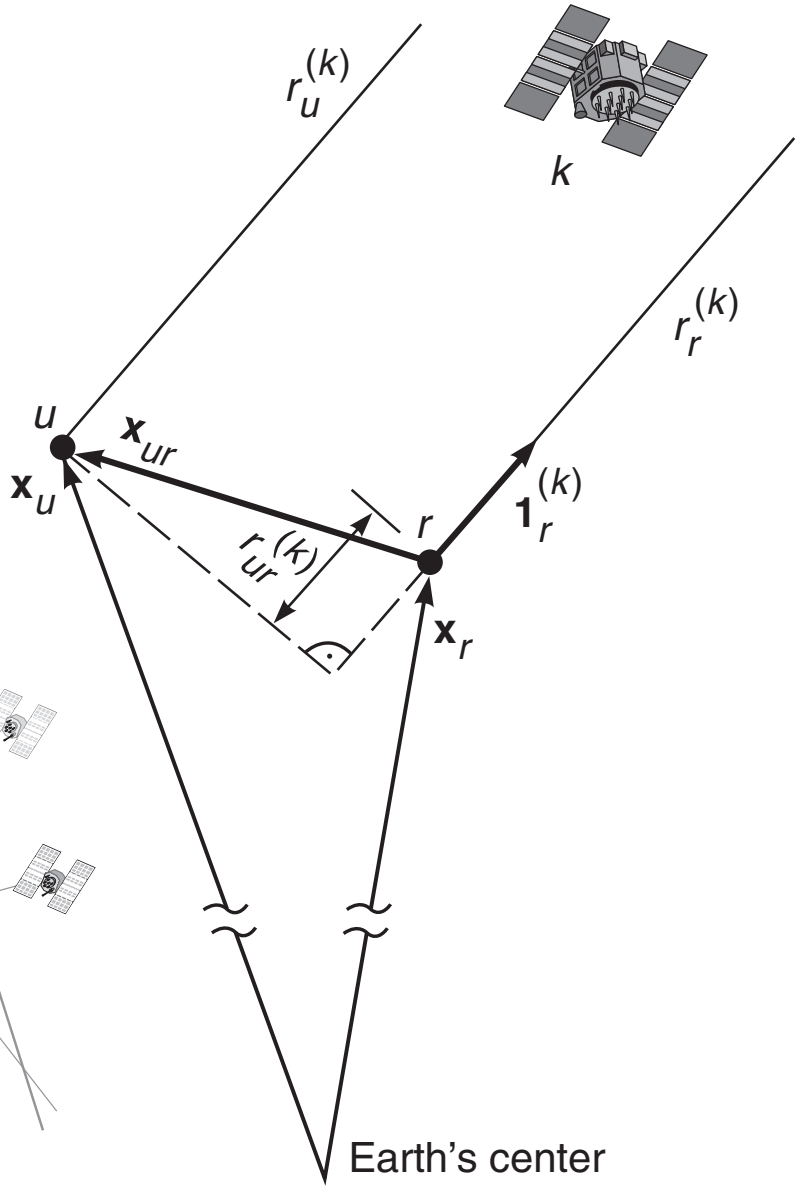
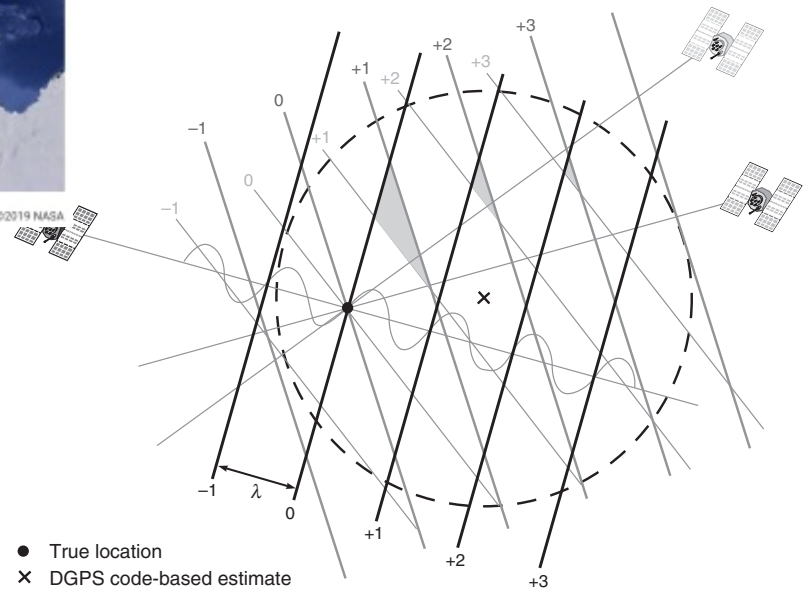
ASEN 5090 Lecture 16
Precise Positioning Methods







Penny Axelrad

Penina.Axelrad@Colorado.edu

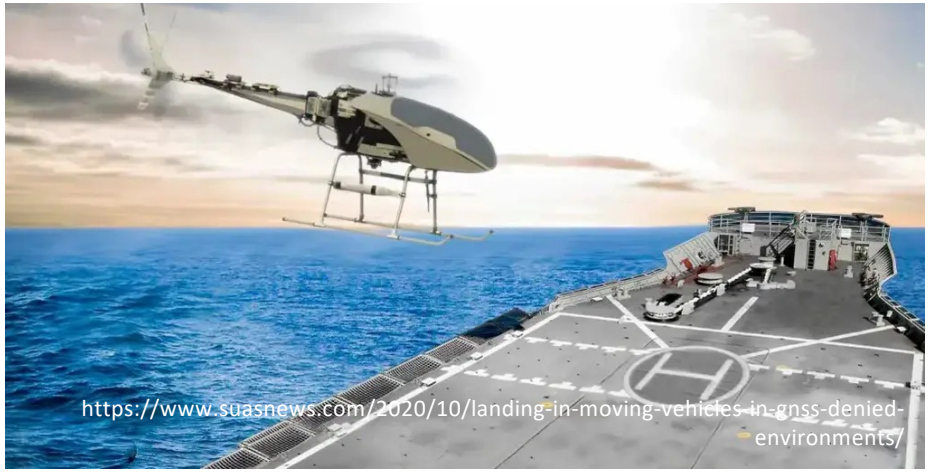
(303) 492-6872

Course schedule

Week	Date	Topic	Reading	Assignments - Due 11pm*
9	10/19/26	L16. Precise Positioning Methods	ME Ch 7	
	10/21/26	Midterm Exam (no lecture)		Midterm Release 10/21 8AM
				Midterm Close 10/25 11PM
10	10/26/26	L17. Introduction to Signals & Spectra	ME Ch 8	
	10/28/26	L18. GNSS Spectrum	ME Ch 9	
11	11/2/26	L19. Receiver Overview	ME Ch 11, PNT Ch 14	HW6 Due, HW7 Assigned
	11/4/26	L20. GNSS Receiver Architecture	ME Ch 10 (parts)	
12	11/9/26	L21. GPS Acquisition	ME Ch 11.2	
	11/11/26	L22. GPS Acquistion (continued)		
	11/13/26			HW7 Due, HW8 Assigned
13	11/16/26	L23. GPS Signal Tracking Basics	ME Ch 12, PNT Ch 15	
	11/18/26	L24. Advanced Signal Tracking (Xu)		

Precise Positioning Methods – carrier phase-based

- Precise Point Positioning (PPP)
- Real-time Kinematic (RTK) Positioning
- Relative Positioning/Navigation

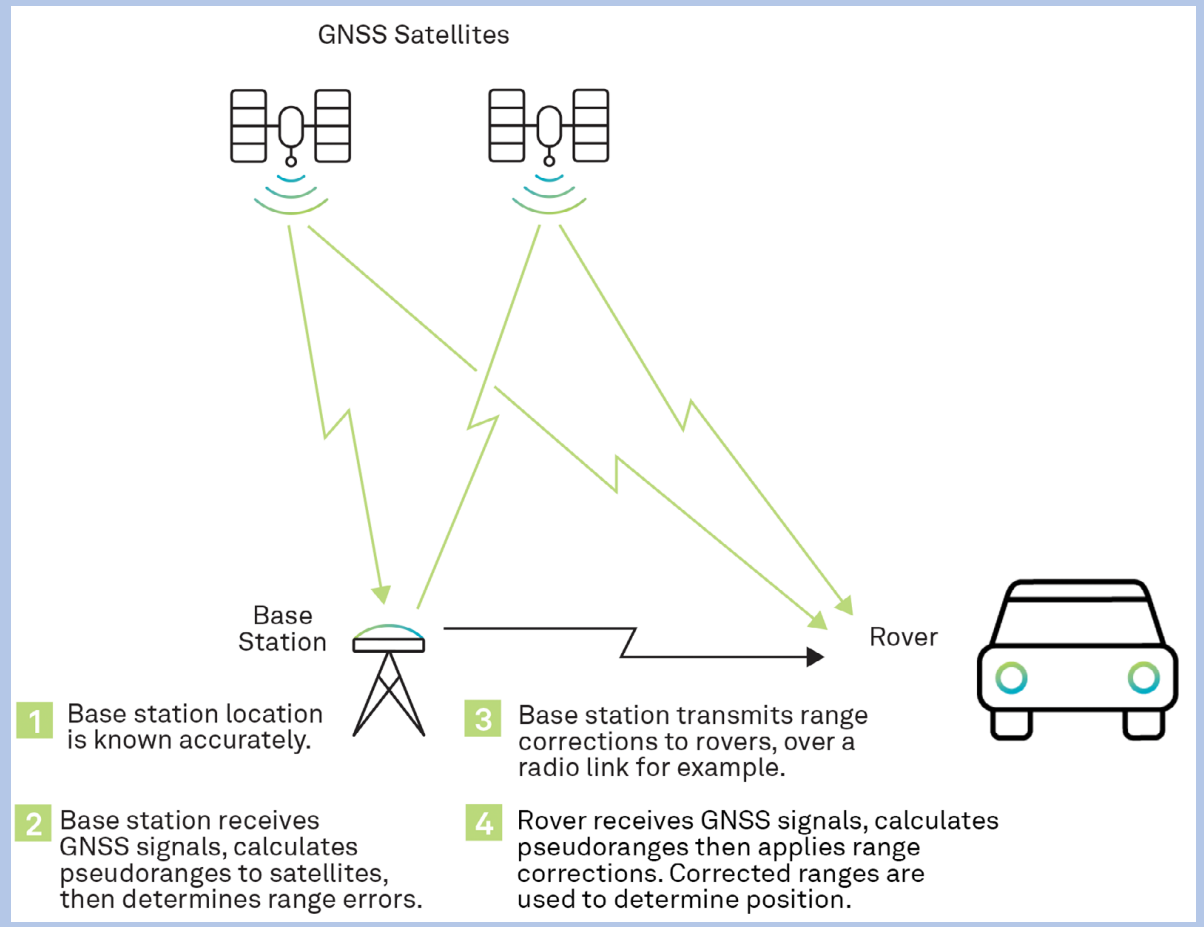


- Challenge: ambiguity resolution
- Strategies:
  - modeling & long-time observations (PPP)
  - differencing (RTK, Relative Navigation)



Positioning Techniques – improve accuracy with local corrections

Differential GPS (DGPS)



Real Time Kinematic (RTK)

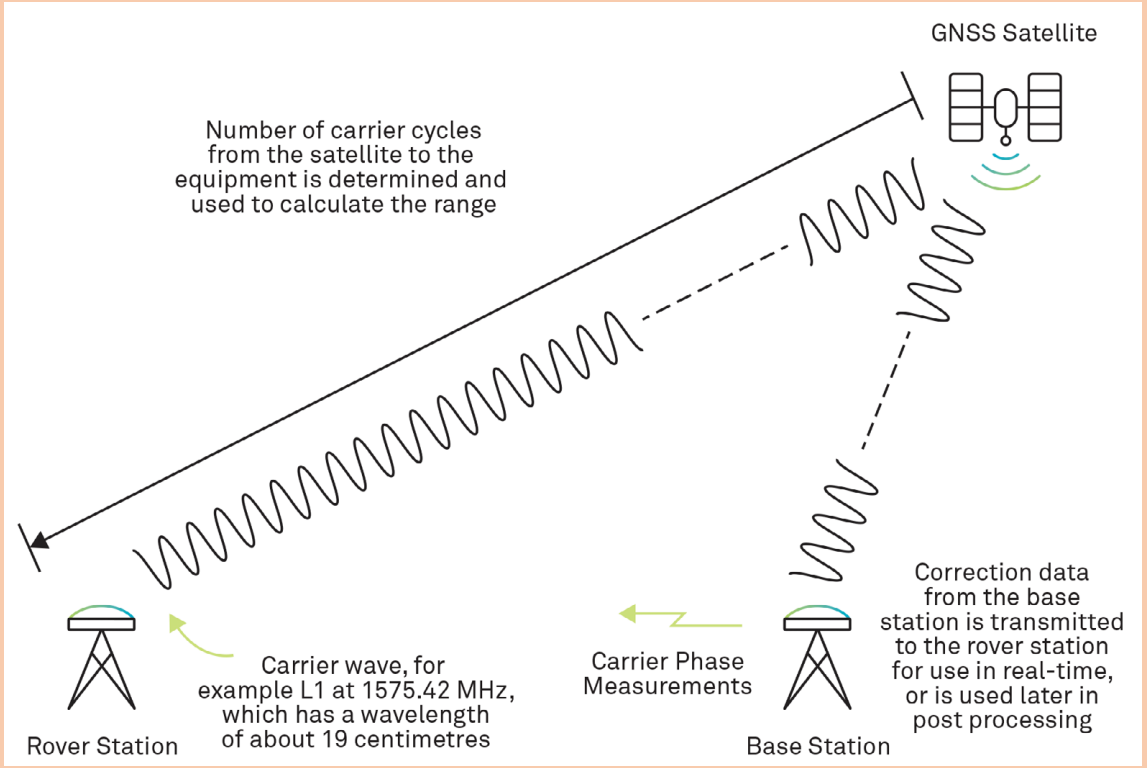
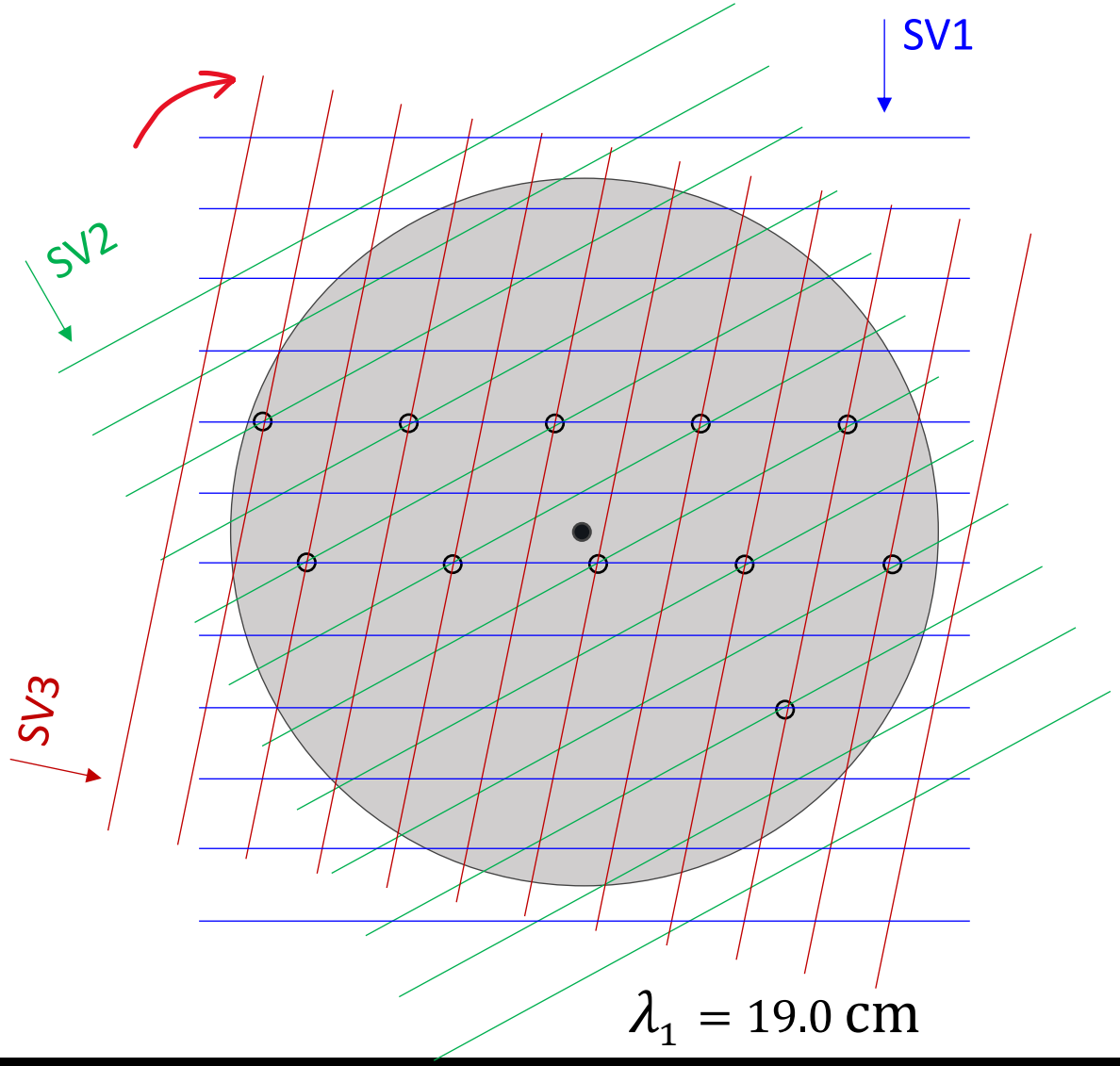


Figure 43 Real-Time Kinematic (RTK) overview

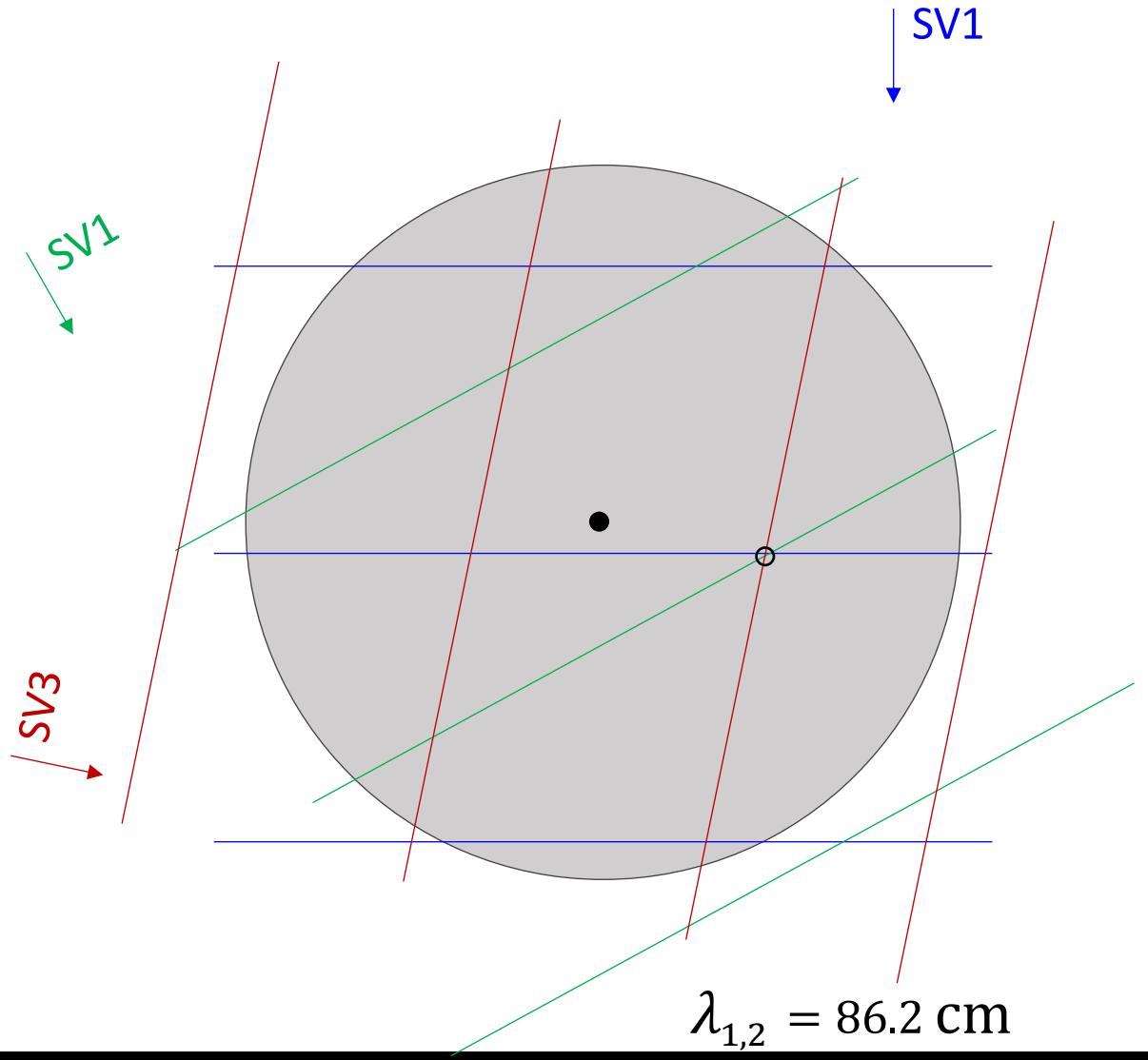
An Introduction to GNSS, Third Edition 53

Attribution: © 2015-2023 NovAtel Inc.

Carrier Phase Measurement Positioning



$\lambda_1 = 19.0\ \text{cm}$



$\lambda_{1,2} = 86.2\ \text{cm}$

What helps the phase ambiguity resolution?

- More satellites
- More frequencies
- Long period of observations

Pseudorange equations as a reference:

$$\rho(t) = r(t, t-\tau) + I_\rho(t) + T_\rho(t) + c\left(\delta t_u(t) - \delta t^s(t-\tau)\right) + \varepsilon_\rho(t)$$

$$\delta\rho^{(s)} = \rho^{(s)} - \rho_0^{(s)} = \begin{bmatrix} -\frac{x^{(s)} - x_0}{R_0^{(s)}} & -\frac{y^{(s)} - y_0}{R_0^{(s)}} & -\frac{z^{(s)} - z_0}{R_0^{(s)}} & 1 \end{bmatrix} \begin{bmatrix} \delta x \\ \delta y \\ \delta z \\ \delta b \end{bmatrix} + \varepsilon^{(s)}$$

Revisit Pseudorange Modeling

Carrier phase smoothing/leveling: be careful with cycle slips

Pseudorange model accuracy/precision: up to ~ 2 decimeters/ ~ 1 meter

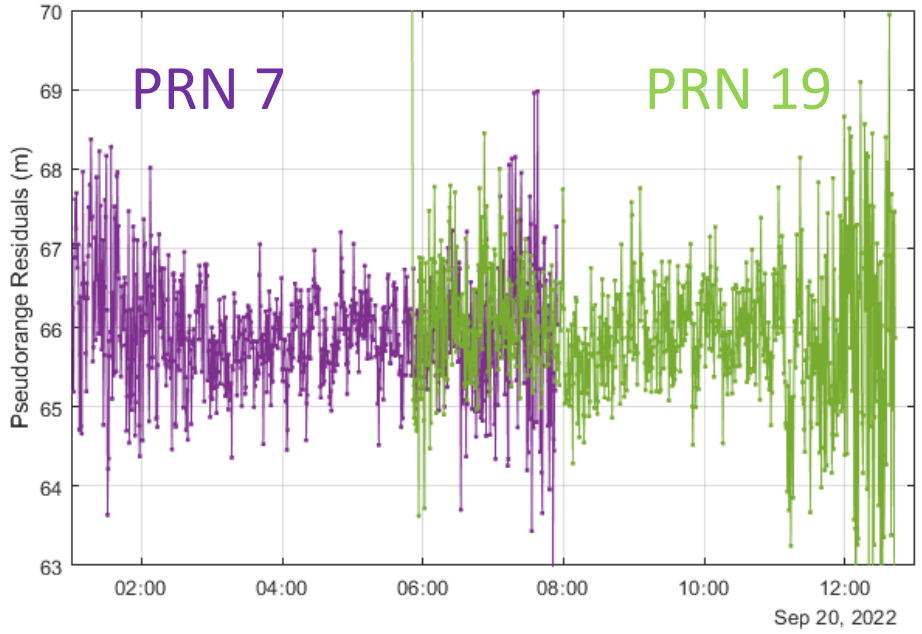
$$\rho(t) = r(t, t-\tau) + I_\rho(t) + T_\rho(t) + c\left(\delta t_u(t) - \delta t^s(t-\tau)\right) + \varepsilon_\rho(t)$$

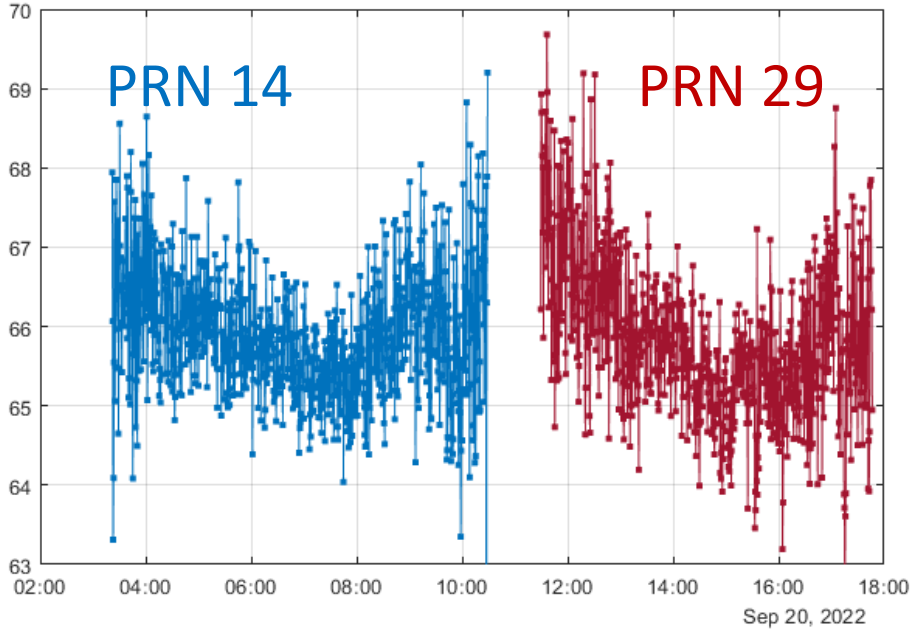
Precise Satellite Mass Center

$$\ldots + \varepsilon_{APO}(t) + \varepsilon_{else}(t)$$

C1W & C2W IF

$$\ldots + \varepsilon_{DCB}(t) + \varepsilon_{APO}(t) + \varepsilon_{else}(t)$$





Carrier Phase Range Modeling

Phase range model:

$$\Phi(t) = r(t, t-\tau) + I_\Phi(t) + T_\Phi(t) + c\left(\delta t_u(t) - \delta t^s(t-\tau)\right) + \varepsilon_\Phi(t) + N\lambda$$

Precise Satellite Mass Center

$$\ldots + \varepsilon_{APO}(t) + \varepsilon_{else}(t) + N\lambda$$

satellite phase bias $\ldots + \varepsilon_{PB}(t) + \varepsilon_{APO}(t) + \varepsilon_{else}(t) + N\lambda$

phase wind-up $\ldots + \varepsilon_{PB}(t) + \varepsilon_{windup}(t) + \varepsilon_{APO}(t) + \varepsilon_{else}(t) + N\lambda$

Precise Point Positioning (PPP)

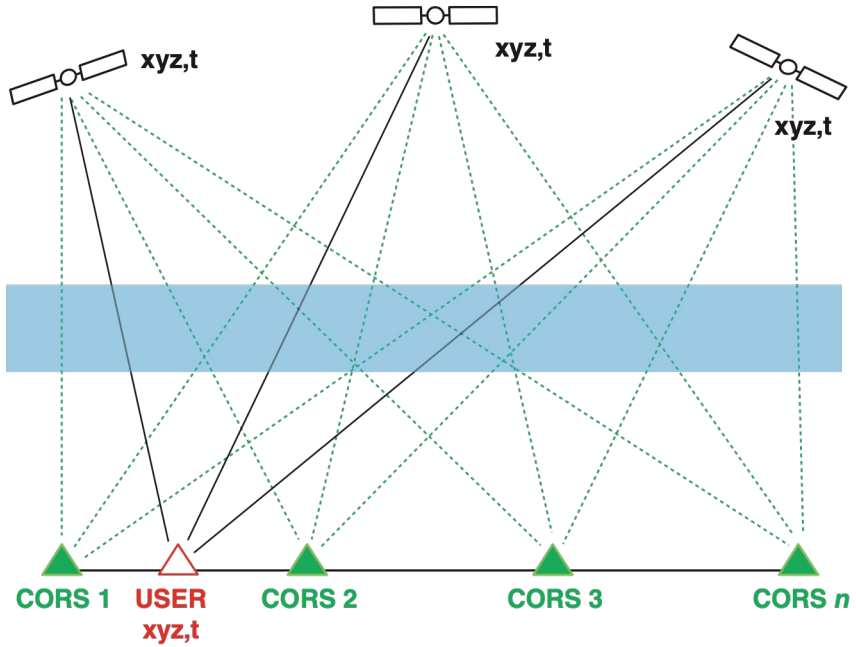


Figure 20.1 Precise Point Positioning (PPP): Single-receiver pseudorange and carrier-phase-based positioning with the aid of externally provided orbit and clock corrections. The precise orbits and clocks are determined by a global network of GNSS-receiver-equipped continuously operating reference stations (CORS).

From P. Teunissen, GNSS Precise Point Positioning, Ch20 in J. Morton et al. PNT Technologies in the 21st Century

- A modeling and processing method with which one can compute positions with high accuracy anywhere on the globe using a single GNSS receiver
- Broadcast orbits and clocks are replaced with precise estimates
- Pseudorange & carrier phase on multiple frequencies

$$p_{u,j}^{st} + dt_j^{st} = \rho_u^{st} + \mu_j \iota_u^{st}$$
$$\phi_{u,j}^{st} + dt_j^{st} = \rho_u^{st} - \mu_j \iota_u^{st} + \lambda_j a_{u,j}^{st} \qquad (20.4)$$

Predicted Products Supporting Real-time PPP

Table 20.3 GPS orbit and clock product table: Broadcast and IGS [5]

GPS		Accuracy (rms)	Latency	Updates	Sample
Orbits and Clocks					**Interval**
Broadcast	Orbits	~100 cm	Real time	—	daily
	Clocks	~5 ns (150 cm)			
Ultra-rapid	Orbits	~5 cm	Real time	Four times	15 min
(predicted)	Clocks	~3 ns (90 cm)		daily	
Ultra-rapid	Orbits	~3 cm	3–9 h	Four times	15 min
(observed)	Clocks	~150 ps (4.5 cm)		daily	
Rapid	Orbits	~2.5 cm	17–41 h	Daily	15 min
	Clocks	~75 ps (2.3 cm)			5 min
Final	Orbits	~2.5 cm	12–18 days	Weekly	15 min
	Clocks	~75 ps (2.3 cm)			30 s

Source: IGS (2015), IGS products, http://igscb.jpl.nasa.gov/components/prods.html. URL http://igscb.jpl.nasa.gov/components/prods.html.

PPP Error Models Summary

- PPP States
  - Correction to user position vector
  - Wet tropo zenith delay
  - Phase ambiguities

Table 20.4 PPP a priori corrections (adapted from [143])

	Correction	Size (approx.)
Satellite	Phase center offset	0.5–3 m
	Phase center variation	5–15 mm
	Clock offset	<1 ms
	Relativistic clock effect	10–20 m
	Differential code biases	Up to 5 m
Atmosphere	Troposphere (dry, vertical)	2.3 m
	Ionosphere (vertical)	Up to 30 m
Receiver	Phase center offset	5–15 cm
	Phase center variation	Up to 3 cm
	Phase wind-up	10 cm
Site movement	Plate motion	Up to 0.1 m/y
	Solid Earth tide	Up to 0.4 m
	Ocean loading	1–10 cm
	Pole tide	25 mm
	Atmospheric loading	—

Source: From Steigenberger, P. (2015) *Accuracy of Current and Future Satellite Navigation Systems, Habilitation Thesis*, Technische Universität München.

PPP – What's the catch?

- Requires access to high precision products
- Requires consistent carrier phase tracking – no cycle slips
- Can require a long time to converge, because measurement sensitivity of many states is similar when the satellite-user geometry changes slowly

Older techniques use single, double, and triple measurement differencing to gain advantages of carrier phase.

- The catch here is that you need to work with observation data from a nearby station.

Single Difference

User receiver carrier phase measurements:

$$\varphi_u^{(k)} = \frac{1}{\lambda}\left(r_u^{(k)} - I_u^{(k)} + T_u^{(k)}\right) + f\left(\delta t_u - \delta t^{(k)}\right) + N_u^{(k)} + \varepsilon_{\varphi,u}^{(k)}$$

Reference (base station) receiver carrier phase measurements:

$$\varphi_r^{(k)} = \frac{1}{\lambda}\left(r_r^{(k)} - I_r^{(k)} + T_r^{(k)}\right) + f\left(\delta t_r - \delta t^{(k)}\right) + N_r^{(k)} + \varepsilon_{\varphi,r}^{(k)}$$

Denote: $x_{ur}^{(k)} = x_u^{(k)} - x_r^{(k)}$

Single difference:

$$\varphi_{ur}^{(k)} = \frac{1}{\lambda}\left(r_{ur}^{(k)} - I_{ur}^{(k)} + T_{ur}^{(k)}\right) + f\delta t_{ur} + N_{ur}^{(k)} + \varepsilon_{\varphi,ur}^{(k)}$$

Short Baseline Approximation

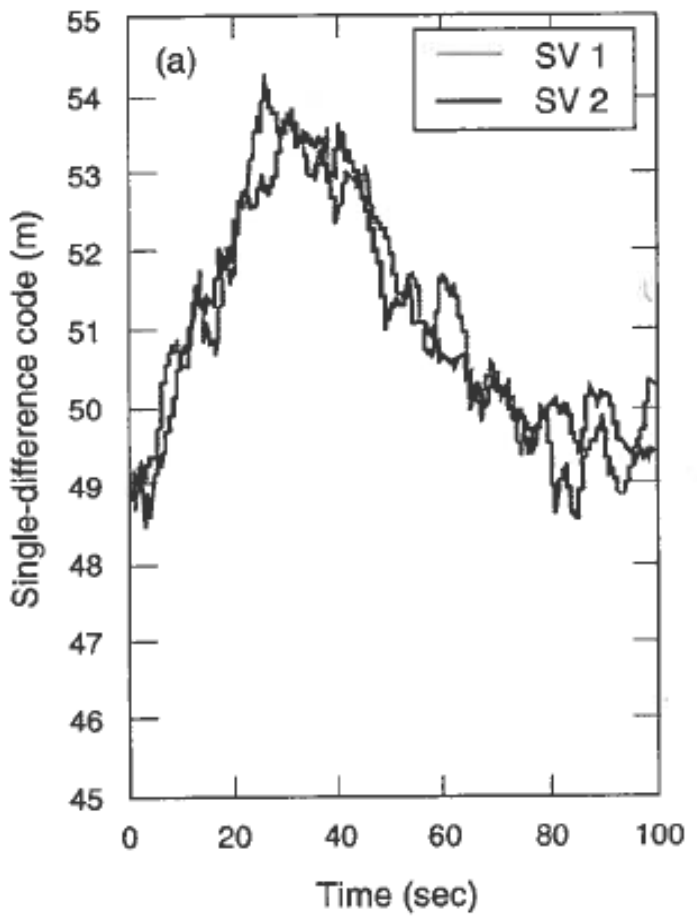
Short baseline: User-reference receiver distance <10km. Iono/tropo are common

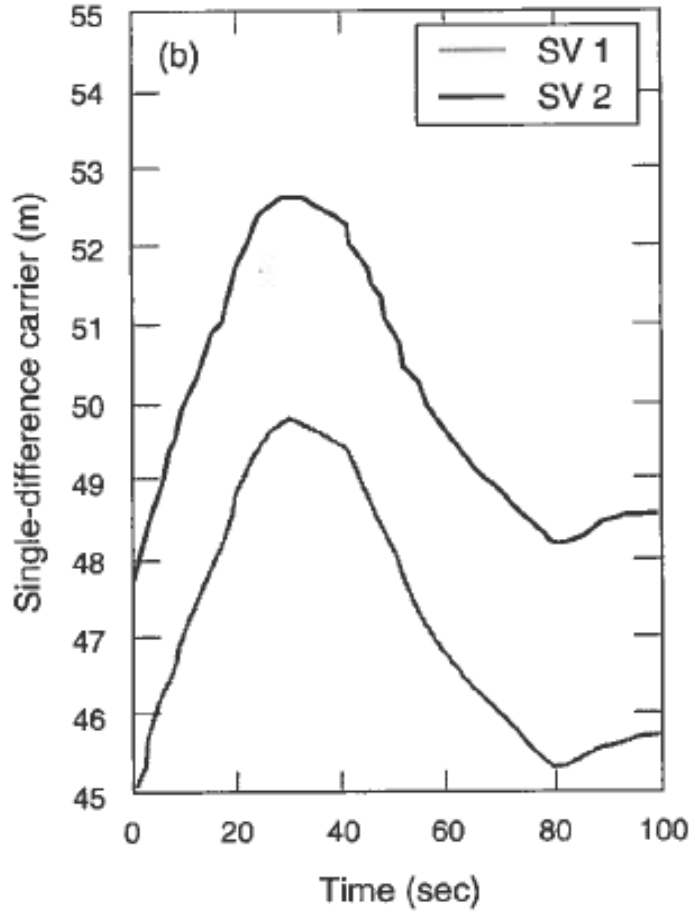
$$\varphi_{ur}^{(k)} \approx \frac{1}{\lambda} r_{ur}^{(k)} + f\delta t_{ur} + N_{ur}^{(k)} + \varepsilon_{\varphi,ur}^{(k)}$$

$$\rho_{ur}^{(k)} \approx r_{ur}^{(k)} + c\delta t_{ur} + \varepsilon_{\rho,ur}^{(k)}$$

Example:
2 stationary RXs
150m baseline
Single difference

Question:
Why do the 4 curves
follow the same pattern?





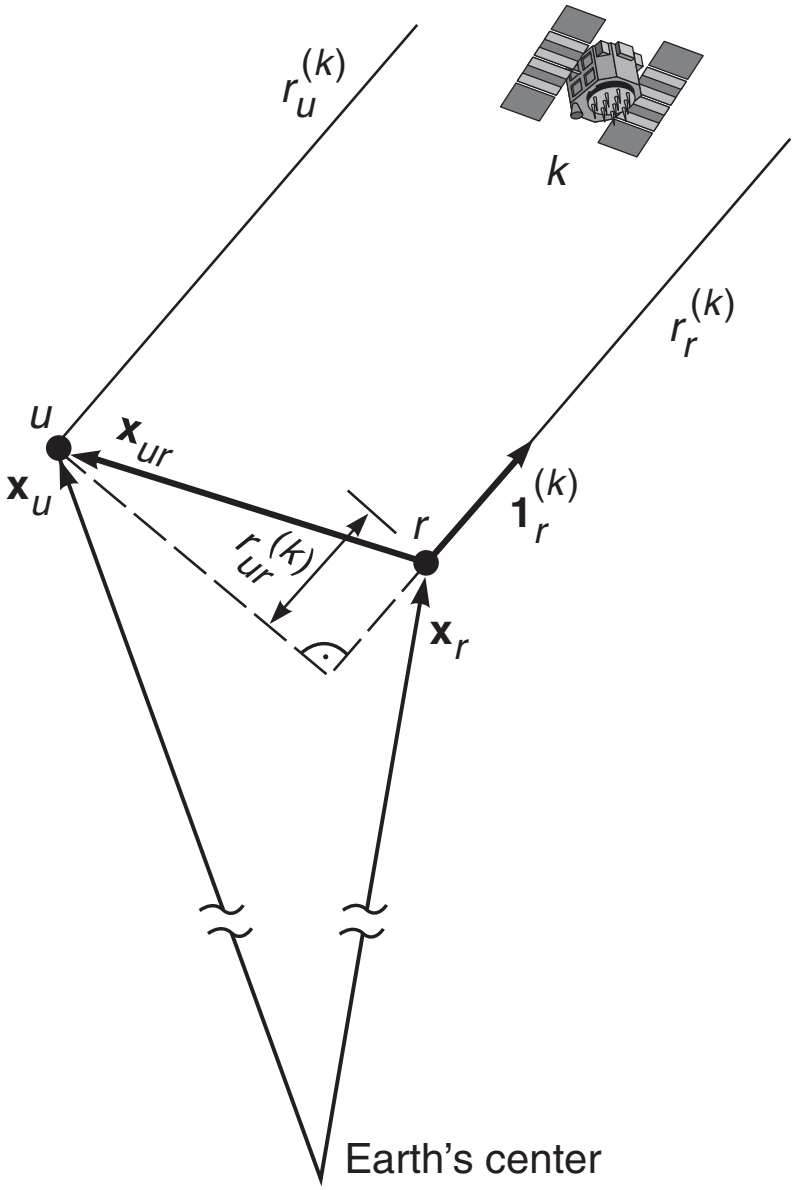
Single Difference Range Equations

$$r_{ur}^{(k)} = r_u^{(k)} - r_r^{(k)} = -\mathbf{l}_r^{(k)} \bullet \mathbf{x}_{ur}$$

$$\phi_{ur}^{(k)} \approx \frac{1}{\lambda}\mathbf{l}_r^{(k)} \bullet \mathbf{x}_{ur} + f\,\delta t_{ur} + N_{ur}^{(k)} + \varepsilon_{\varphi,ur}^{(k)}$$

For K SVs,

$$\boldsymbol{\phi}_{ur} = \lambda^{-1}\begin{bmatrix} (-\mathbf{1}_r^{(1)})^T \\ (-\mathbf{1}_r^{(2)})^T \\ \vdots \\ (-\mathbf{1}_r^{(K)})^T \end{bmatrix}\mathbf{x}_{ur} + f \cdot \delta t_{ur}\begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} + \begin{bmatrix} N_{ur}^{(1)} \\ N_{ur}^{(2)} \\ \vdots \\ N_{ur}^{(K)} \end{bmatrix} + \boldsymbol{\varepsilon}_{\phi,ur}$$



ME Figure 7.4

More on Single Difference Range Equations

Define: $\Phi_{ur}^{(k)} = \lambda\varphi_{ur}^{(k)} \qquad b_{ur} = c\delta t_{ur} = f\lambda\delta t_{ur}$

$$\mathbf{\Phi}_{ur} = \underbrace{\begin{bmatrix} \left(-\mathbf{1}_r^{(1)}\right)^T & 1 \\ \left(-\mathbf{1}_r^{(2)}\right)^T & 1 \\ \vdots & \vdots \\ \left(-\mathbf{1}_r^{(K)}\right)^T & 1 \end{bmatrix}}_{\mathbf{G}_{ur}} \underbrace{\begin{bmatrix} \mathbf{x}_{ur} \\ b_{ur} + \lambda N_{ur}^{(1)} \end{bmatrix}}_{\text{Relative user-reference position}} + \underbrace{\begin{bmatrix} 0 \\ \lambda\left(N_{ur}^{(2)} - N_{ur}^{(1)}\right) \\ \vdots \\ \lambda\left(N_{ur}^{(K)} - N_{ur}^{(1)}\right) \end{bmatrix}}_{\boldsymbol{\delta}\mathbf{N}_{ur}} + \boldsymbol{\varepsilon}_{\Phi,ur}$$

Measurement

$\boldsymbol{\delta}\mathbf{N}_{ur}$: Obtained during initialization

$$\mathbf{\Phi}_{ur} - \boldsymbol{\delta}\mathbf{N}_{ur} = \mathbf{G}_{ur}\begin{bmatrix} \mathbf{x}_{ur} \\ b_{ur} + \lambda N_{ur}^{(1)} \end{bmatrix} + \boldsymbol{\varepsilon}_{\boldsymbol{\Phi},\boldsymbol{ur}}$$

Relative Position Solution

$$\underbrace{\boldsymbol{\Phi}_{ur} - \boldsymbol{\delta}\mathbf{N}_{ur}}_{\boldsymbol{\delta\Phi}_{ur}} = \mathbf{G}_{ur}\begin{bmatrix}\mathbf{x}_{ur}\\ b_{ur} + \lambda N_{ur}^{(1)}\end{bmatrix} + \boldsymbol{\varepsilon}_{\boldsymbol{\Phi},ur}$$

$$\boldsymbol{\delta\Phi}_{ur} = \mathbf{G}_{ur}\begin{bmatrix}\mathbf{x}_{ur}\\ b_{ur} + \lambda N_{ur}^{(1)}\end{bmatrix} + \boldsymbol{\varepsilon}_{\boldsymbol{\Phi},ur}$$

Compare with single receiver linearized range equations:

$$\delta\boldsymbol{\rho} = \mathbf{G}\begin{bmatrix}\delta\mathbf{x}\\ \delta b\end{bmatrix} + \tilde{\boldsymbol{\varepsilon}}_{\rho} \quad \Rightarrow \quad \begin{bmatrix}\delta\hat{\mathbf{x}}\\ \delta\hat{b}\end{bmatrix} = \left(\mathbf{G}^T\mathbf{G}\right)^{-1}\mathbf{G}^T\delta\boldsymbol{\rho}$$

We can obtain relative position estimation:

$$\begin{bmatrix}\mathbf{x}_{ur}\\ b_{ur} + \lambda N_{ur}^{(1)}\end{bmatrix} = \left(\mathbf{G}_{\boldsymbol{ur}}^{\boldsymbol{T}}\mathbf{G}_{ur}\right)^{-1}\mathbf{G}_{\boldsymbol{ur}}^{\boldsymbol{T}}\boldsymbol{\delta\Phi}_{ur}$$

Double Difference

Single difference for SV *k* and *l*:

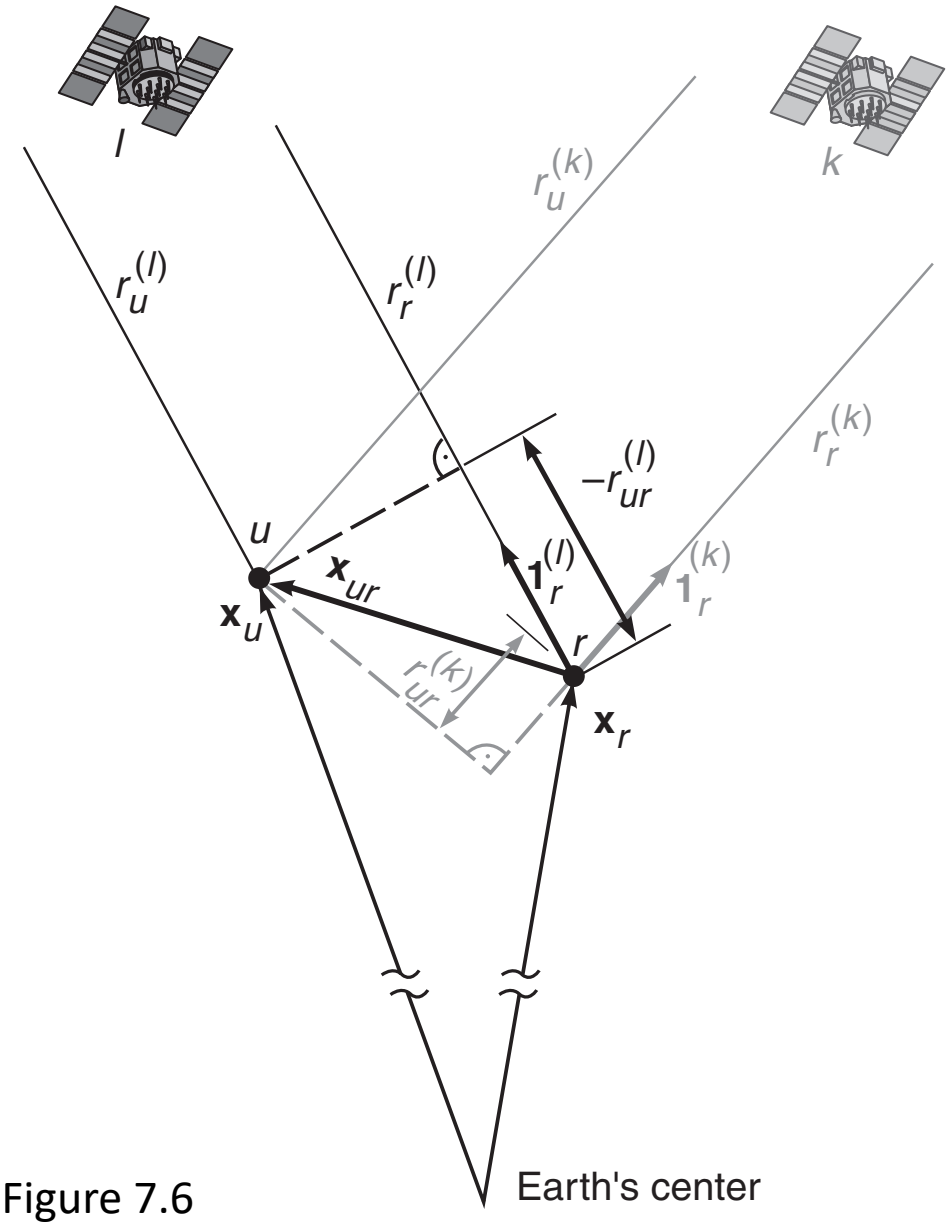
$$\phi_{ur}^{(k)} = \frac{1}{\lambda}\left(r_{ur}^{(k)} - I_{ur}^{(k)} + T_{ur}^{(k)}\right) + f\,\delta t_{ur} + N_{ur}^{(k)} + \varepsilon_{\varphi,ur}^{(k)}$$

$$\phi_{ur}^{(l)} = \frac{1}{\lambda}\left(r_{ur}^{(l)} - I_{ur}^{(l)} + T_{ur}^{(l)}\right) + f\,\delta t_{ur} + N_{ur}^{(l)} + \varepsilon_{\varphi,ur}^{(l)}$$

Double difference of single differences:

$$\phi_{ur}^{(kl)} = \phi_{ur}^{(k)} - \phi_{ur}^{(l)} = \left(\phi_{u}^{(k)} - \phi_{r}^{(k)}\right) - \left(\phi_{u}^{(l)} - \phi_{r}^{(l)}\right)$$

$$\phi_{ur}^{(kl)} = \frac{1}{\lambda}\left(r_{ur}^{(kl)} - I_{ur}^{(kl)} + T_{ur}^{(kl)}\right) + N_{ur}^{(kl)} + \varepsilon_{\varphi,ur}^{(kl)}$$



ME Figure 7.6

Short Baseline Double Difference

For a short baseline, iono/tropo are common: $\phi_{ur}^{(kl)} = \frac{1}{\lambda} r_{ur}^{(kl)} + N_{ur}^{(kl)} + \varepsilon_{\varphi,ur}^{(kl)}$

$$r_{ur}^{(kl)} = (r_u^{(k)} - r_r^{(k)}) - (r_u^{(l)} - r_r^{(l)}) = -(\mathbf{1}_r^{(k)} - \mathbf{1}_r^{(l)}) \cdot \mathbf{x}_{ur}$$

Double difference over K SVs:

$$\begin{bmatrix} \phi_{ur}^{(21)} \\ \phi_{ur}^{(31)} \\ \vdots \\ \phi_{ur}^{(K1)} \end{bmatrix} = \lambda^{-1} \begin{bmatrix} -\left(\mathbf{1}_r^{(2)} - \mathbf{1}_r^{(1)}\right)^T \\ -\left(\mathbf{1}_r^{(3)} - \mathbf{1}_r^{(1)}\right)^T \\ \vdots \\ -\left(\mathbf{1}_r^{(K)} - \mathbf{1}_r^{(1)}\right)^T \end{bmatrix} \mathbf{x}_{ur} + \begin{bmatrix} N_{ur}^{(21)} \\ N_{ur}^{(31)} \\ \vdots \\ N_{ur}^{(K1)} \end{bmatrix} + \begin{bmatrix} \varepsilon_{\phi,ur}^{(21)} \\ \varepsilon_{\phi,ur}^{(31)} \\ \vdots \\ \varepsilon_{\phi,ur}^{(K1)} \end{bmatrix}$$

Taking the same approach as single difference:

$$\mathbf{\Phi}_{ur}^{kl} = \mathbf{G}_{ur}^{kl}\mathbf{x}_{ur} + \boldsymbol{\varepsilon}_{ur}^{kl} \qquad \mathbf{x}_{ur} = \left(\left(\boldsymbol{G}_{ur}^{kl}\right)^T \boldsymbol{G}_{ur}^{kl}\right)^{-1} \left(\boldsymbol{G}_{ur}^{kl}\right)^T \mathbf{\Phi}_{ur}^{kl}$$

Triple Difference

$$\phi_{ur}^{(kl)} = \frac{1}{\lambda} r_{ur}^{(kl)} + N_{ur}^{(kl)} + \varepsilon_{\varphi,ur}^{(kl)}$$

$$\delta\phi_{ur}^{(kl)}(i) = \phi_{ur}^{(kl)}(t_{i+1}) - \phi_{ur}^{(kl)}(t_i) = \lambda^{-1}\delta r_{ur}^{(kl)}(i) + \delta\varepsilon_{\phi,ur}^{(kl)}(i)$$

$$\delta\boldsymbol{\phi} = \lambda^{-1}\underbrace{\begin{bmatrix} -\delta\left(\mathbf{1}_r^{(2)} - \mathbf{1}_r^{(1)}\right)^T \\ -\delta\left(\mathbf{1}_r^{(3)} - \mathbf{1}_r^{(1)}\right)^T \\ \vdots \\ -\delta\left(\mathbf{1}_r^{(K)} - \mathbf{1}_r^{(1)}\right)^T \end{bmatrix}}_{\text{Large DOP}} \mathbf{x} + \tilde{\boldsymbol{\varepsilon}}$$

Advantage: You do not have to resolve the carrier ambiguity

Resolve Integer Ambiguity: Using Pseudorange Measurements

Double carrier phase difference: $\phi_{ur}^{(kl)} = \frac{1}{\lambda} r_{ur}^{(kl)} + N_{ur}^{(kl)} + \varepsilon_{\varphi,ur}^{(kl)}$

Short notation: $\phi_q = \frac{1}{\lambda} r + N_q + \varepsilon_{\phi,q}$ (q=L_1, or L_2)

Double pseudorange difference: $\rho_q = r + \varepsilon_{\rho,q}$

$$\hat{N}_q = \left[\phi_q - \frac{\rho_q}{\lambda_q} \right]_{roundoff}$$

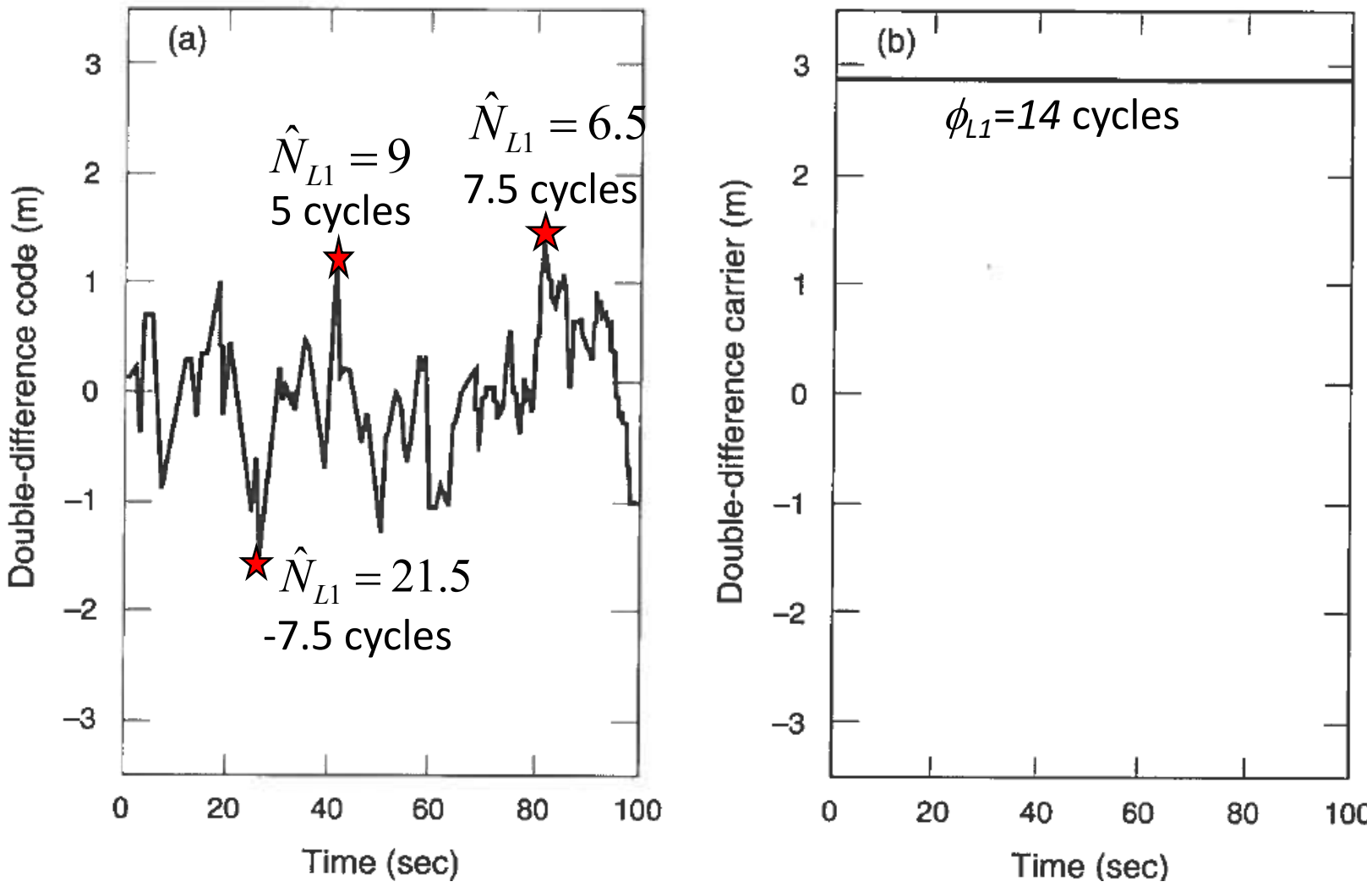
$$\sigma(\phi_{L1}) \approx 0.05 cycles \qquad \sigma(\rho_{L1}) \approx 1m \sim 5 cycles \qquad \sigma(N_{L1}) \sim 5 cycles$$

Averaging over time to reduce the integer estimation error

Example: 150 m Baseline Measurements

$$\hat{N}_{L1} = \left[\phi_{L1} - \frac{\rho_{L1}}{\lambda_{L1}} \right]_{roundoff}$$

$$E\left\{ \hat{N}_{L1} \right\} = 15$$



Integer Estimate for L1 Double-difference Measurements

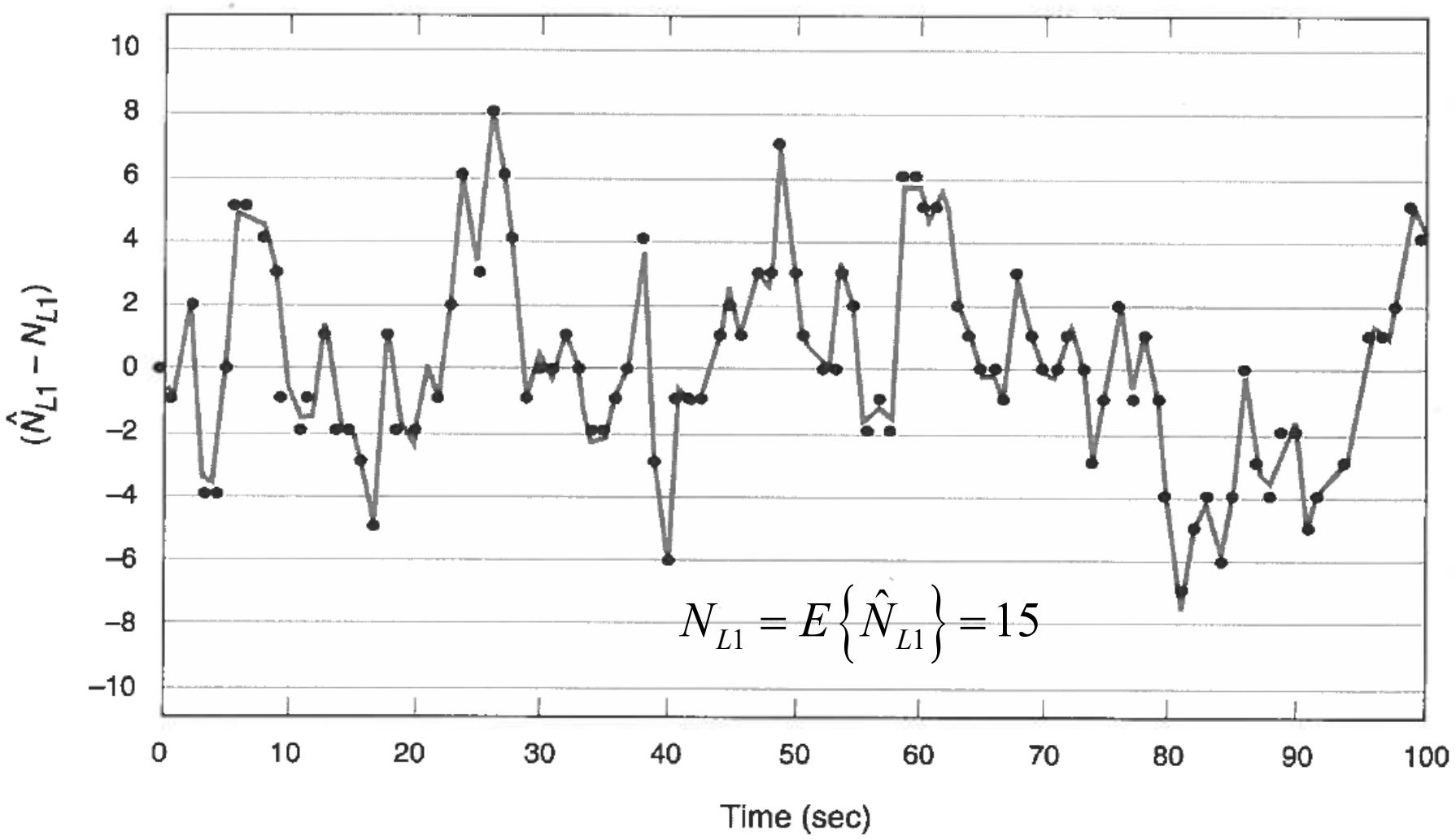


Figure 7.7 Integer estimates for L1 double-difference measurements from the 150-meter baseline. (Courtesy of Dr. Jaewoo Jung, Trimble Navigation)

Dual-Frequency Measurements: Wide-Lane

$$\widehat{N}_q = \left[\phi_q - \frac{\rho_q}{\lambda_q}\right]_{roundoff}$$

Larger λ_q → Smaller number of code cycles → Better estimates

Create a "beat" signal with a larger wavelength:

$$\phi_{L12} = \phi_{L1} - \phi_{L2} = r\left(\frac{1}{\lambda_{L1}} - \frac{1}{\lambda_{L2}}\right) + (N_{L1} - N_{L2}) + \varepsilon_{\phi_{L12}}$$

$$= r\left(\frac{f_{L1} - f_{L2}}{c}\right) + (N_{L1} - N_{L2}) + \varepsilon_{\phi_{L12}}$$

$$= \frac{r}{\lambda_{L12}} + N_{L12} + \varepsilon_{\phi_{L12}}$$

$\lambda_{L12} = c/(f_{L1} - f_{L2}) = 0.862$ m →

Wide-lane double difference integer ambiguity: $\hat{N}_{L12} = \left[\phi_{L12} - \frac{\rho_{L1}}{\lambda_{L12}}\right]_{roundoff}$

Example: 150m Baseline Wide-Lane Integer Ambiguity

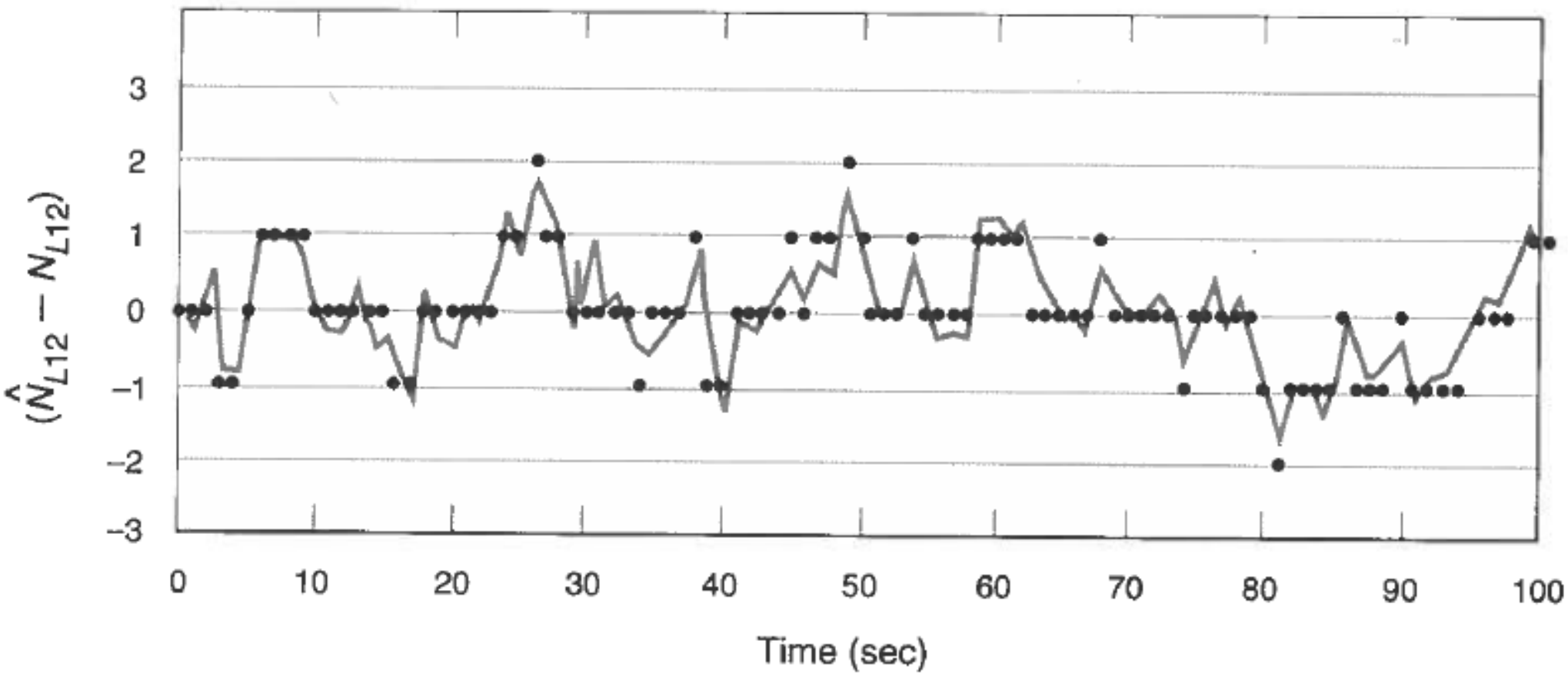


Figure 7.8 Integer estimates for the wide-lane double-difference measurements from the 150-meter baseline. (Courtesy of Dr. Jaewoo Jung, Trimble Navigation)

Conversion from Wide-Lane Integer to L1 Integer

$$\phi_{L1} = \frac{r}{\lambda_{L1}} + N_{L1} + \tilde{\varepsilon}_{\phi_{L1}}$$

$$\phi_{L2} = \frac{r}{\lambda_{L2}} + N_{L2} + \tilde{\varepsilon}_{\phi_{L2}}$$

$$N_{L1} - \frac{\lambda_{L2}}{\lambda_{L1}} N_{L2} = \phi_{L1} - \frac{\lambda_{L2}}{\lambda_{L1}} \phi_{L2} + \varepsilon$$

$$N_{L1} - N_{L2} = N_{L12}$$

Solve for N_{L1} and N_{L2}

$$\hat{N}_{L1} = \left(\frac{\lambda_{L2}}{\lambda_{L1}} - 1 \right)^{-1} \left(\frac{\lambda_{L2}}{\lambda_{L1}} N_{L12} - \phi_{L1} + \frac{\lambda_{L2}}{\lambda_{L1}} \phi_{L2} \right)$$

Wide-Lane Integer Ambiguity

Table 7.2. Carriers and beat-frequency signals to be available when GPS Modernization is completed

Carrier Signal	*Frequency (MHz)*	*Wavelength (m)*
L1	$f_{L1} = 1575.42$	$\lambda_{L1} = 0.190$
L2	$f_{L2} = 1227.60$	$\lambda_{L2} = 0.244$
L5	$f_{L5} = 1176.45$	$\lambda_{L5} = 0.255$
L1–L5	$f_{L15} = 398.97$	$\lambda_{L15} = 0.751$ (medium lane)
L1–L2	$f_{L12} = 347.82$	$\lambda_{L12} = 0.862$ (wide lane)
L2–L5	$f_{L25} = 51.15$	$\lambda_{L25} = 5.861$ (extra-wide lane)

Integer Ambiguity Resolution

Algorithms:

Least-squares Ambiguity Decorrelation Adjustment (LAMBDA)

(Teunissen, 1996; Teunissen et al. 1997)

Algorithms:

Local Minima Search (LMS)

(Pratt et al. 1997)

$$\boldsymbol{y = G\delta x + N + \varepsilon}$$

Find $\boldsymbol{\delta x}$ and $\boldsymbol{N}$ that minimize cost function:

$$\boldsymbol{C_W(\delta x, N) = \|y - G\delta x - N\|_W^2}$$

$$\boldsymbol{= (y - G\delta x - N)^T W (y - G\delta x - N)}$$

Reading Assignment

- Misra & Enge Textbook Ch 7

- References ：
- P. Teunissen, GNSS Precise Point Positioning, Ch20 in J. Morton et al. PNT Technologies in the 21st Century
- LAMBDA Paper: https://link.springer.com/article/10.1007/BF00863419
- Link to article about biases in code and carrier due to transmitter and receiver: https://rdcu.be/b8IBP

- **Useful links:**
- https://www.rtklib.com/
- https://gnss.curtin.edu.au/research/lambda-and-ps-lambda-software-packages/

Acknowledgements

ASEN5090 lecture materials have been developed at the University of Colorado Boulder by Professors Penina Axelrad, Dennis Akos, Kristine M. Larson, Jade Morton, and Yang Wang, with additional contributions by many outstanding TA's and students.

Slides in this lecture are primarily from Prof Morton and Prof Wang